# SMART FAKE NEWS DETECTION SYSTEM

## Major Project Report

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## 1. Introduction

The rapid spread of information through websites and social media has made it easier for people to access news. At the same time, false or misleading information can circulate widely, create confusion and influence public opinion.

This project presents a machine-learning-based Smart Fake News Detection System that classifies a news item as Fake News or Real News based on its textual content. It applies Natural Language Processing (NLP), converts text into numerical features using TF-IDF, and compares Logistic Regression with Multinomial Naive Bayes. A Streamlit application provides an interface for entering a headline and article and viewing the model's prediction.

## 2. Problem Statement

Identifying misleading news manually can be difficult and time-consuming, particularly when large amounts of information are shared online. The aim is to develop a text-classification system that analyses a news headline and article and predicts whether the supplied text belongs to the fake-news or real-news category.

The system is an educational machine-learning application. Its output is a prediction based on patterns learned from the selected dataset; it is not independent verification of the facts or sources in a news article.

## 3. Objectives

Develop a machine-learning system that classifies news text into Fake News and Real News categories.

Apply text-cleaning and normalisation techniques to prepare news headlines and articles.

Convert textual data into numerical features using TF-IDF.

Train and compare Logistic Regression and Multinomial Naive Bayes classifiers.

Evaluate model performance using accuracy, precision, recall and F1-score.

Build a simple Streamlit interface for submitting news text and viewing predictions.

## 4. Dataset Description

The project uses the ISOT Fake and Real News Dataset, obtained from Kaggle. It is supplied in two CSV files: Fake.csv, containing records labelled as fake news, and True.csv, containing records labelled as real news.

The original combined dataset contained 44,898 records. After duplicate records and records with empty titles or article text were removed, 44,058 records remained. The cleaned dataset contains the fields title, text, subject, date and label.

The project assigns label 0 to Fake News and label 1 to Real News. The cleaned dataset contains 22,848 fakenews records and 21,210 real-news records.

Dataset source: https://www.kaggle.com/datasets/clmentbisaillon/fake-and-real-news-dataset

Dataset reuse and redistribution should follow the terms provided by the dataset publisher and Kaggle. The dataset licence should be checked before redistributing the original CSV files.

## 5. Data Preprocessing

Both CSV files were loaded using Pandas, and a label column was added to identify each record. The datasets were combined into one DataFrame. Completely empty Unnamed columns were removed, duplicate records were removed, and records with missing or empty titles or article text were excluded.

The title and text fields were converted to strings and leading and trailing whitespace was removed. The headline and article text were combined into one content field. The combined content was converted to lowercase; URLs and characters other than English letters and whitespace were removed; repeated whitespace was reduced to a single space.

## 6. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the structure and characteristics of the cleaned dataset before model training. Four visualisations were generated.

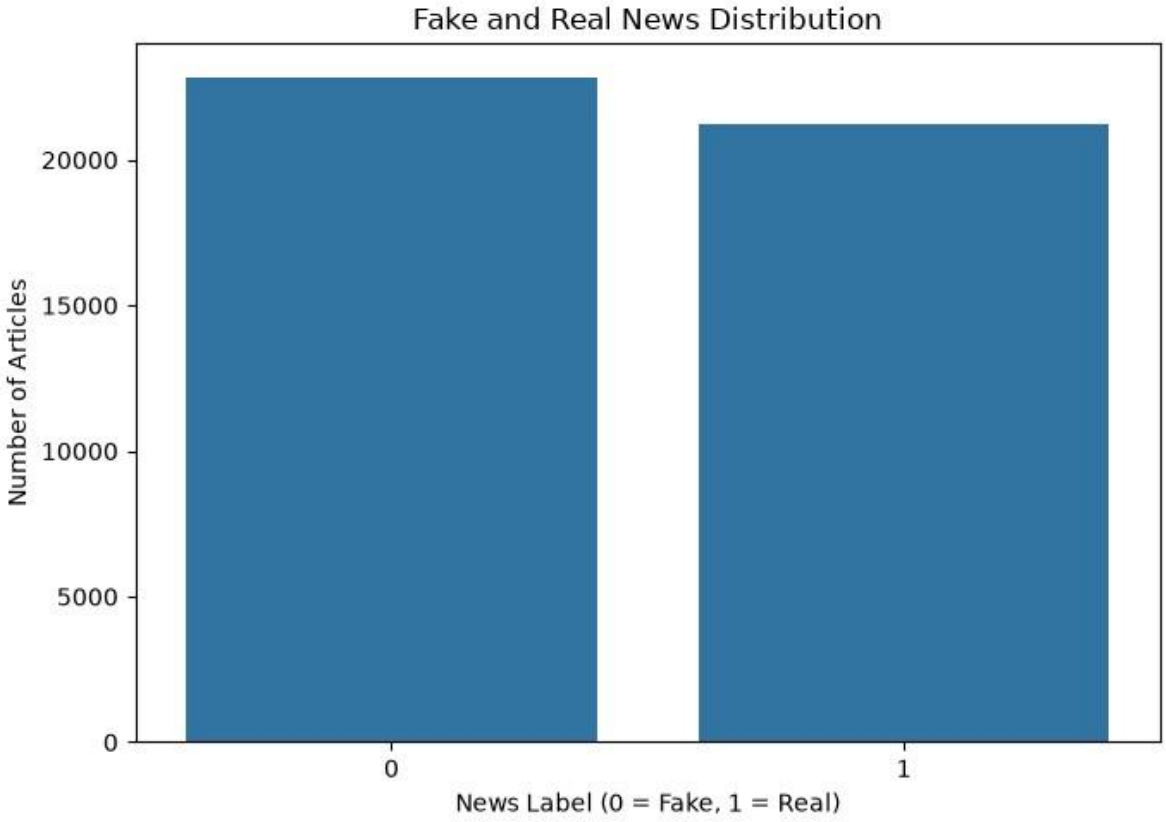



**Figure 1. Label Distribution**

The bar chart compares the number of fake-news and real-news records in the cleaned dataset. It shows that both classes are represented in substantial numbers, with 22,848 fake-news records and 21,210 real-news records.

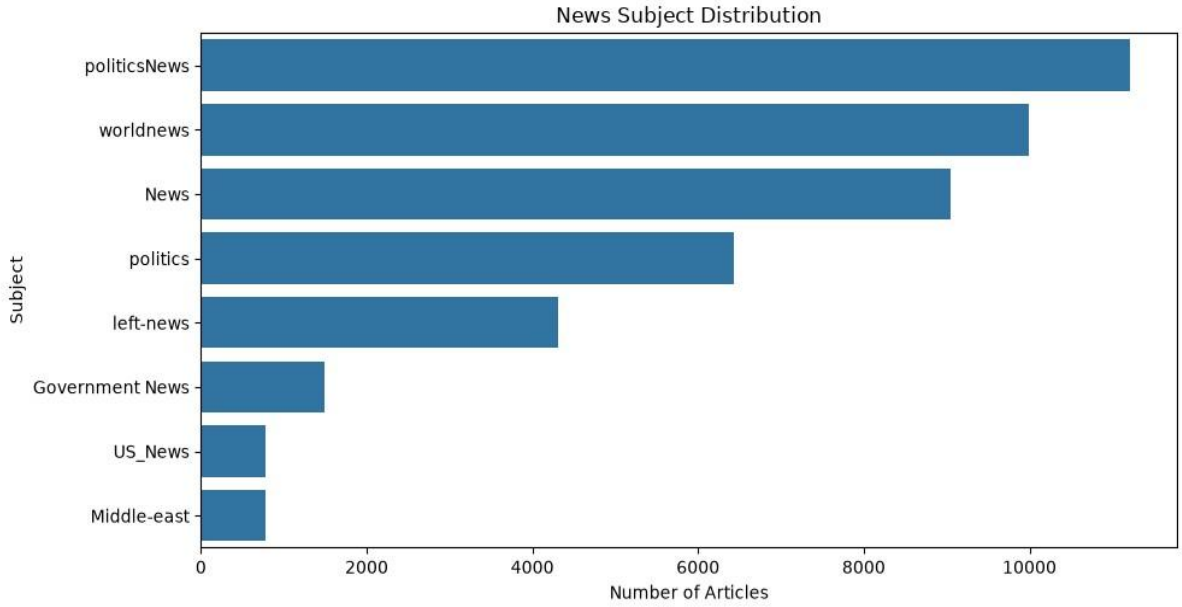



**Figure 2. Subject Distribution**

The horizontal bar chart displays the number of records associated with each news subject and distinguishes the fake and real categories. It helps show which subject categories are most represented in the dataset.

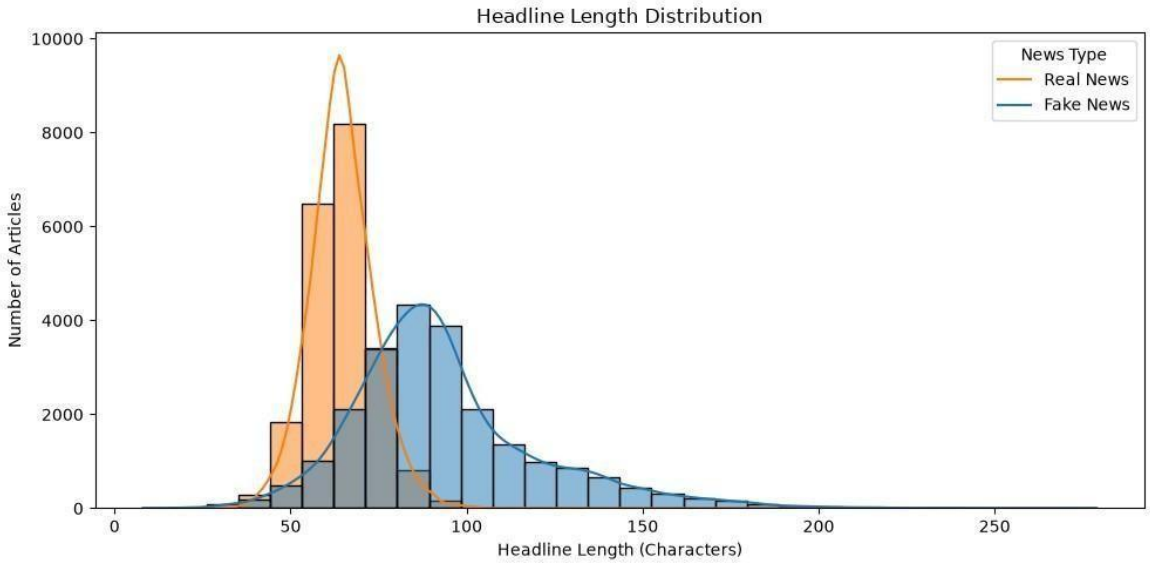



**Figure 3. Headline Length Distribution**

The histogram shows the distribution of headline lengths, measured in characters, for fake and real news records. It illustrates variation in headline size and the overlap between the two classes.

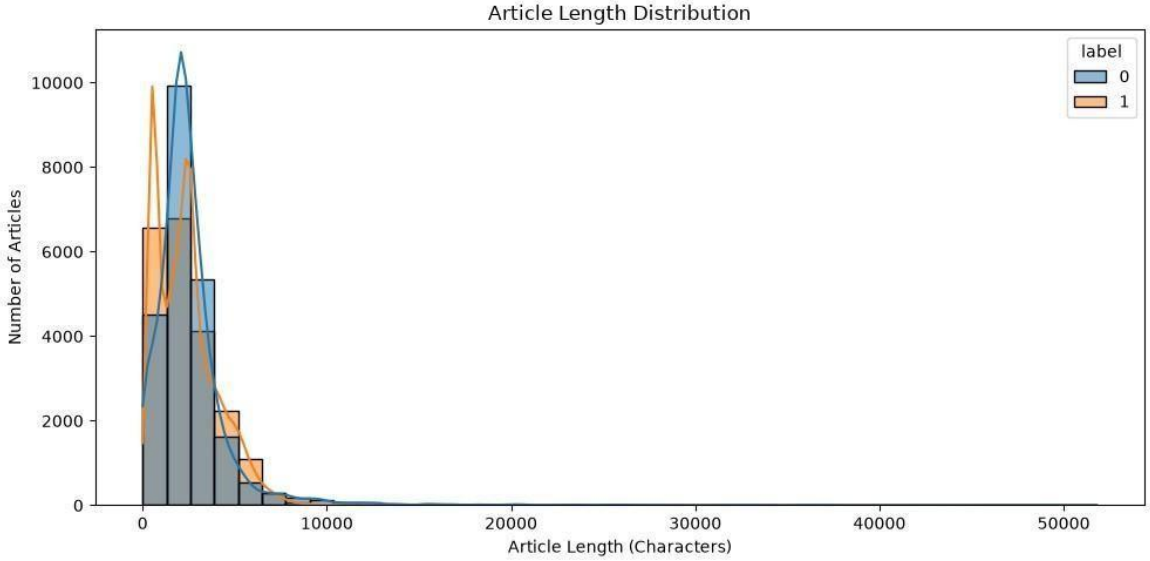



**Figure 4. Article Length Distribution**

The histogram shows the distribution of article lengths, measured in characters, for the two labels. It illustrates that article text varies considerably in length, with many shorter articles and a smaller number of much longer articles.

## 7. Feature Extraction Using TF-IDF

Machine-learning algorithms require numerical input, so the prepared text was converted into numerical features using Term Frequency–Inverse Document Frequency (TF-IDF). TF-IDF gives weight to terms according to their frequency in a document and their distribution across the collection of documents.

The vectorizer was configured with a maximum of 5,000 features, English stop-word removal and an n-gram range of (1, 2), allowing it to represent individual words and two-word sequences. The vectorizer was fitted only on the training data and then used to transform the test data. This prevents information from the test set being used when learning the vocabulary and feature weights.

## 8. Machine Learning Algorithms

Logistic Regression is a supervised learning algorithm commonly used for classification. It estimates the relationship between input features and class membership and predicts one of the defined categories. In this project, it classifies news text as Fake News or Real News.

Multinomial Naive Bayes is a probabilistic classification algorithm frequently used for text-classification tasks. It applies Bayes' theorem and models the contribution of features to each class. It was trained on the same TFIDF training features to provide a comparison with Logistic Regression.

Both algorithms used the same training and test split so their reported test results could be compared on the same data.

## 9. Model Training and Testing

The cleaned dataset was divided into training and testing subsets using an 80:20 split. A stratified split preserved approximately the same class proportions in both subsets. The split contained 35,246 training records and 8,812 testing records.

The TF-IDF vectorizer was fitted on the training text and used to transform both the training and testing text. Logistic Regression and Multinomial Naive Bayes were trained on the training features. Their predictions were generated for the held-out test features.

## 10. Model Evaluation

Accuracy measures the proportion of predictions that were correct. Precision measures the proportion of predicted positive cases that were correct. Recall measures the proportion of actual positive cases that were identified correctly. The F1-score combines precision and recall into a single measure.

The table reports macro-averaged precision, recall and F1-score, which calculate the metric for each class and then give both classes equal weight.

| Metric | Logistic Regression | Multinomial Naive Bayes |
|---|---|---|
| Accuracy | 99.10% | 94.36% |
| Macro Precision | 99.09% | 94.35% |
| Macro Recall | 99.11% | 94.36% |
| Macro F1-score | 99.10% | 94.35% |

Logistic Regression achieved the highest macro F1-score and was selected as the final model for the application.

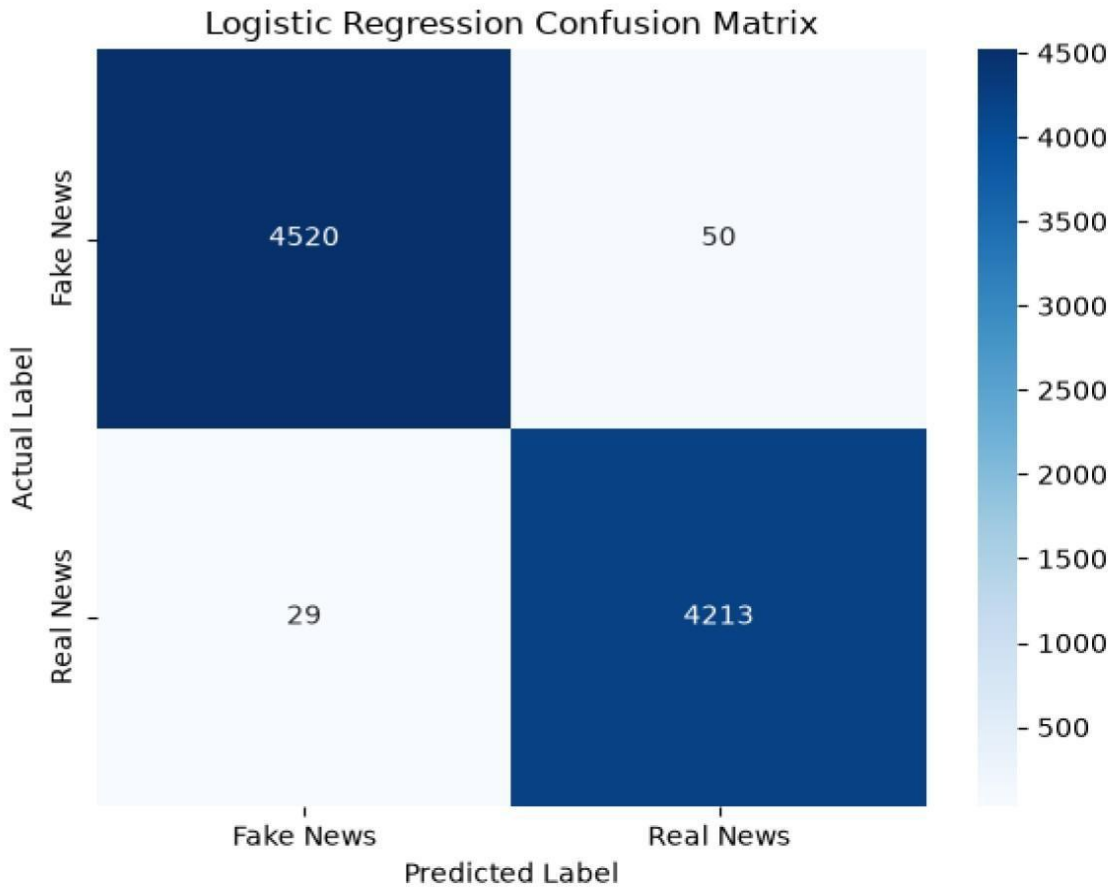



**Figure 5. Logistic Regression Confusion Matrix**

The confusion matrix compares actual and predicted labels for Logistic Regression. The diagonal cells show correct classifications: 4,520 fake-news records and 4,213 real-news records. The off-diagonal cells show 50 fake records predicted as real and 29 real records predicted as fake.

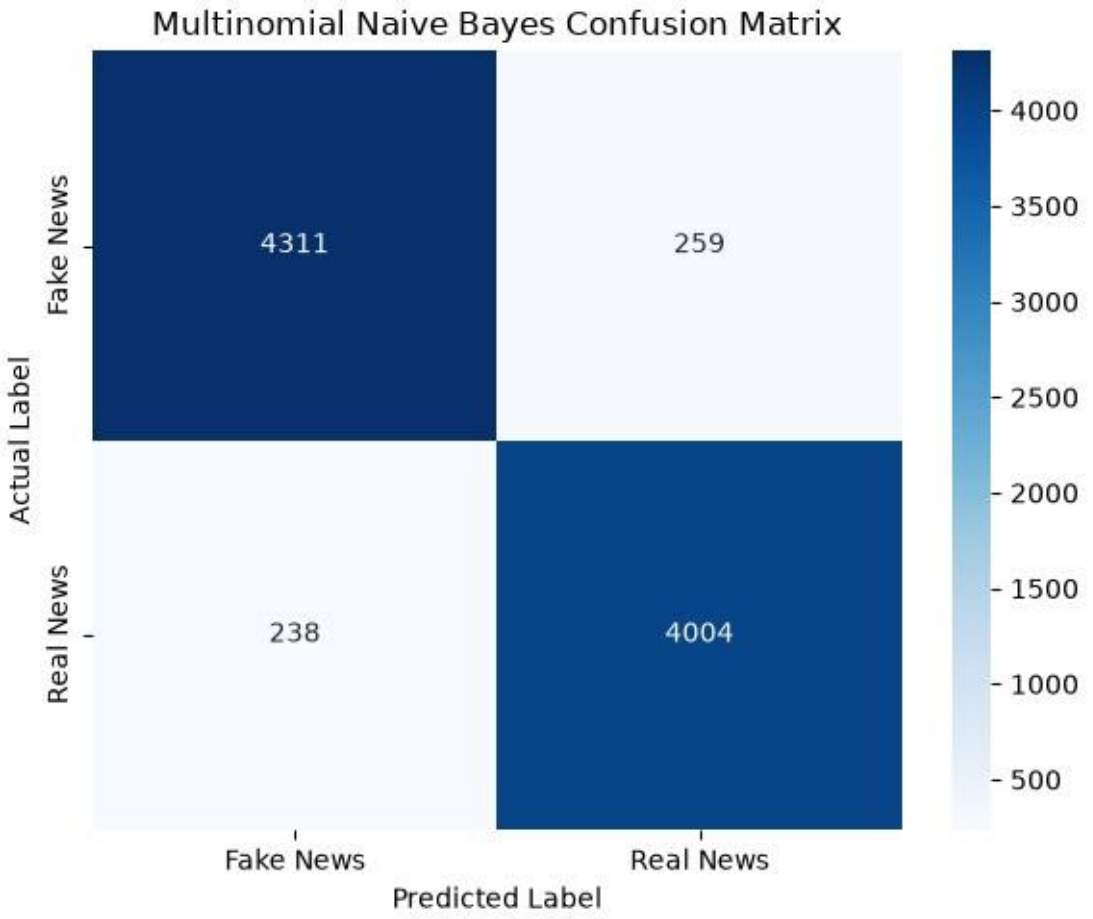



**Figure 6. Multinomial Naive Bayes Confusion Matrix**

The confusion matrix compares actual and predicted labels for Multinomial Naive Bayes. The diagonal cells show 4,311 correctly classified fake-news records and 4,004 correctly classified real-news records. The off-diagonal cells show 259 fake records predicted as real and 238 real records predicted as fake.

## 11. Application Development

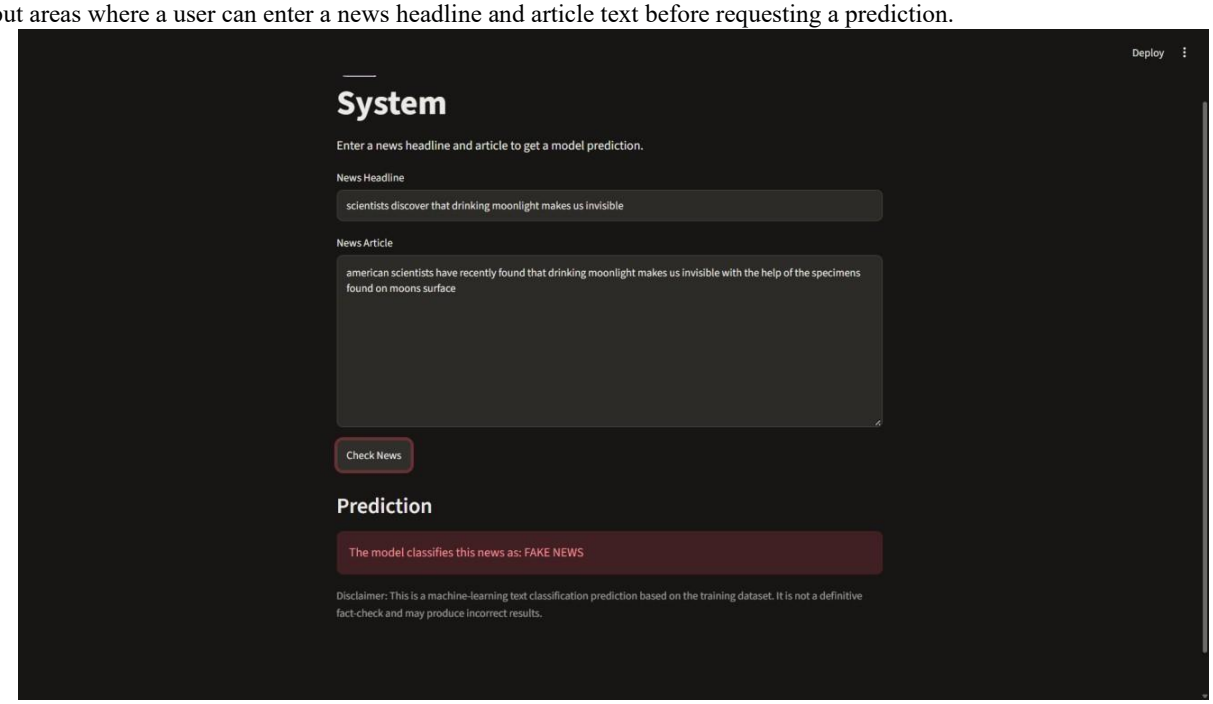
A web interface was developed using Streamlit. It allows a user to enter a news headline and article text and submit the text for classification. The application loads the saved model and fitted TF-IDF vectorizer using Joblib rather than retraining the models each time a prediction is requested.

When the user submits text, the application applies the same text-normalisation steps used during training, transforms the text with the saved TF-IDF vectorizer and passes the resulting features to the saved classification model. The application then displays the predicted category as Fake News or Real News. It also warns that the output is a model prediction and not a definitive fact-check.



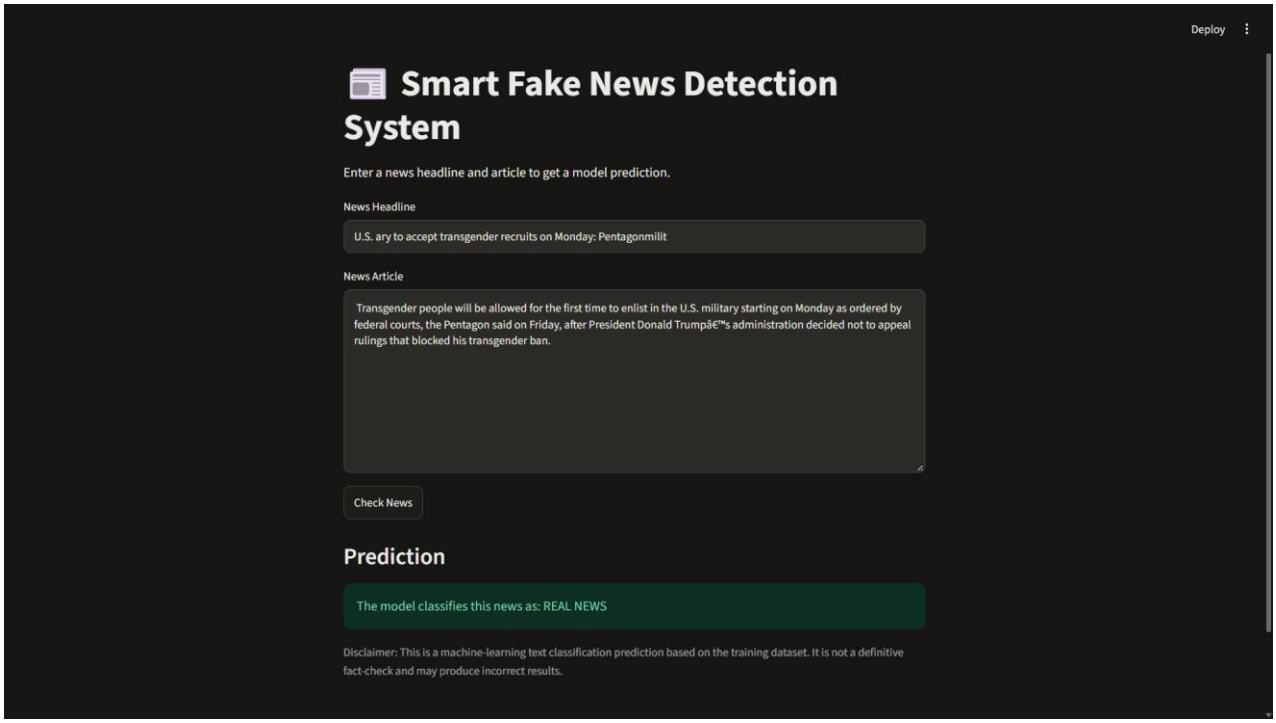

**Screenshot 1. Streamlit Application Interface**

This screenshot shows the main interface of the Smart Fake News Detection System. It presents the application title and the input areas where a user can enter a news headline and article text before requesting a prediction.




**Screenshot 2. Fake News Prediction**

This screenshot shows an example prediction in which the application displays the result as Fake News after the submitted text is processed by the saved model and TF-IDF vectorizer. It demonstrates the prediction workflow; it is not proof that every article labelled fake is factually false.




**Screenshot 3. Real News Prediction**

This screenshot shows an example prediction in which the application displays the result as Real News. It demonstrates the application's real-news output for the submitted sample; the result remains a model prediction rather than independent fact verification.

## 12. Results and Discussion

On the held-out test set, Logistic Regression achieved an accuracy of 99.10%, while Multinomial Naive Bayes achieved an accuracy of 94.36%. Logistic Regression also achieved the higher macro-averaged precision, recall and F1-score. It was therefore selected as the model used by the Streamlit application.

The results indicate that Logistic Regression performed better than Multinomial Naive Bayes on this particular test split. However, a high score on one dataset does not establish that the system will perform equally well on new sources, different topics or current events. Dataset-specific patterns, possible source-related signals and similarities between records can affect evaluation results. The reported scores should be interpreted as test performance on the selected dataset, not as a guarantee of real-world fact-checking accuracy.

## 13. Limitations and Future Scope

The system depends on the quality, coverage and labelling of the dataset used for training. Patterns present in the dataset may not represent all types of news or all publishers. The model may make incorrect predictions for satire, breaking news, unfamiliar subjects or articles whose writing style differs from the training examples. The application analyses text patterns and does not independently verify claims, consult authoritative sources or establish the factual accuracy of an article.

Future improvements could include evaluating the system on more diverse and recently collected datasets, investigating duplicate and near-duplicate records more thoroughly, and testing performance on articles from sources not represented in the training data. The system could also be extended with source-credibility information and links to independent fact-checking resources. Further work could explore transformer-based NLP models and deployment as an online service.

## 14. Conclusion

The Smart Fake News Detection System demonstrates an end-to-end text-classification workflow using Natural Language Processing and machine learning. The project involved dataset loading and cleaning, exploratory data analysis, text normalisation, TF-IDF feature extraction, model training, evaluation and development of a Streamlit interface.

Logistic Regression achieved better results than Multinomial Naive Bayes on the held-out test set and was selected for use in the application. The completed application accepts a headline and article text and displays the model's predicted category. The project provides practical experience in building and evaluating a machinelearning application while illustrating the importance of interpreting model predictions carefully.

## 15. References


1. ISOT Fake and Real News Dataset. Kaggle. https://www.kaggle.com/datasets/clmentbisaillon/fake-and-realnews-dataset
2. Scikit-learn documentation. https://scikit-learn.org/
3. Pandas documentation. https://pandas.pydata.org/docs/
4. Streamlit documentation. https://docs.streamlit.io/
5. Python documentation. https://docs.python.org/3/